

AIM:

TO write a assembly language program to perform
BLOCK COMPARE operating by pointer concept using
BOBSMP KIT

APPARATUS REQUIRED:

1. MICROPROCESSOR KIT
2. power supply.

ALGORITHM:

1. Start the program.
2. Initialize 05 to the 'c'-register. And load the 'hc' & 'be'-register pair to the accumulator to its specified address.
3. Move 60 to 'b'-register, And load the 'd'-register to Accumulator.
4. Compare the Accumulator values between the memory values.
5. Check the conditions, like Increment (or) Decrement
6. If there is no zero, then jump loop. store the result
7. Halt the program.



Assembly Language Program Chart

Expt. Name : BLOCK COMPARE

Exp.	Address	Label	Mnemonics		Machine Code	Comments
			Opcode	Operand		
	8000		MVI	C, 05	0E, 05	Initialize the 'C' to the '5'
	8002		LXI H	9000	21, 00, 90	Load the HL-pair in the 9000 address
	8005		LXI D	9100	11, 00, 91	Load the DE-pair in the 9100 address.
	8008		MVI	B, 00	06, 00	Move 00 to the 'B'-register
	800A		LDA	XD	1A	Load the register 'D' data
	800B		CMP	M	BE	Compare the memory and Accumulator
	800C		JNC	loop1	92, 10, 80	Jump to loop1, if there is no carry
	800F		INR	B	04	Increment the 'B'-reg
	8010		INX	H	23	Increment the HL-pair
	8011		INX	D	13	Increment the 'D'-reg
	8012		DCR	C	0D	Decrement the 'C'-reg
	8013		JNZ	loop2	C2, 0A, 80	jump to loop2, if there is no zero
	8016		MOV	A, B	7B	move B reg to Acc.
	8017		STA	9150	92, 50, 91	store the result in the 9150 address
	801A		HLT		76	stop the program.

TABULATION:

BEFORE EXECUTION			AFTER EXECUTION		
ADDRESS	INPUT	ADDRESS	INPUT	ADDRESS	OUTPUT
9000	22	9100	11		
9001	33	9101	23		
9002	44	9102	55		
9003	66	9103	44		
9004	77	9104	33	9150	03

Expt. Name

Address La

8000

8002

8005

8008

800A

800B

800C

800F

8010

8011

8012

8013

8016

8017

801A

M. Singh 25/03.

RESULT:

thus the assembly language program for perform
BLOCK OPERATIONS using pointer concept was successfully
executed & output was verified.

